

--	--	--	--	--

Centre Number

--	--	--	--	--	--	--	--	--	--

Student Number

2021

Mathematics Extension 2

Trial HSC Examination

Date: Monday 23th August, 2021

General Instructions

- Reading time – 10 minutes
- Working time – 3 hours
- Write using blue or black pen
- NESA approved calculators may be used
- Show relevant mathematical reasoning and/or calculations

Total Marks:
100

Section I – 10 marks

- Allow about 15 minutes for this section

Section II – 90 marks

- Allow about 2 hours and 45 minutes for this section

Section I (10 marks)	Multiple Choice	/10
Section II (90 marks)	Question 11	/15
	Question 12	/15
	Question 13	/13
	Question 14	/17
	Question 15	/15
	Question 16	/15
Total		/100

This question paper must not be removed from the examination room.

This assessment task constitutes 0% of the course.

BLANK PAGE

Section I

10 marks

Allow about 15 minutes for this section

Use the multiple-choice sheet for Question 1–10

1. If $z_1 = \sqrt{3} + i\sqrt{3}$ and $z_2 = \sqrt{3} + i$, then the complex number $\left(\frac{z_1}{z_2}\right)$ lies in the quadrant
- A. I
- B. II
- C. III
- D. IV
2. The resolved part of the force $\vec{F} = \vec{i} + 2\vec{j} - 4\vec{k}$ in the direction of $\vec{a} = 2\vec{i} + 4\vec{j} - 4\vec{k}$ is:
- A. $\frac{13}{9}(-\vec{i} + 2\vec{j} - 2\vec{k})$
- B. $\frac{13}{9}(\vec{i} + 2\vec{j} - 2\vec{k})$
- C. $\frac{13}{9}(-\vec{i} - 2\vec{j} - 2\vec{k})$
- D. $\frac{26}{3}(\vec{i} + 2\vec{j} - 2\vec{k})$
3. The direction cosines of a line are $\left(\frac{1}{c}, \frac{1}{c}, \frac{1}{c}\right)$, then
- A. $c > 0$
- B. $0 < c < 1$
- C. $c = \pm\sqrt{3}$
- D. $c > 2$

Section I continues

4. The period of the function $f(x) = |\sin x| + |\cos x|$ is:
- A. $\frac{\pi}{2}$
- B. π
- C. $\frac{3\pi}{2}$
- D. 2π
5. The solutions to $z^n = 1 + i$, $n \in \mathcal{Z}^+$ are given by
- A. $2^{\frac{1}{2n}} e^{i(\frac{\pi}{4n} + \frac{2\pi k}{n})}$, $k \in \mathcal{R}$
- B. $2^{\frac{1}{n}} e^{i(\frac{\pi}{4n} + 2\pi k)}$, $k \in \mathcal{Z}$
- C. $2^{\frac{1}{2n}} e^{i(\frac{\pi}{4} + \frac{2\pi k}{n})}$, $k \in \mathcal{Z}$
- D. $2^{\frac{1}{2n}} e^{i(\frac{\pi}{4n} + \frac{2\pi k}{n})}$, $k \in \mathcal{Z}$
6. If $f(y) = e^y$, $g(y) = y$; $y \geq 0$ and
- $$F(t) = \int_0^t f(t-y)g(y)dy, \quad \text{then}$$
- A. $F(t) = t e^{-t}$
- B. $F(t) = 1 - e^{-t}(1+t)$
- C. $F(t) = e^t - (1+t)$
- D. $F(t) = t e^{-t}$

Section I continues

7. Which of the following statements is a negation of the following statement?

$$x \in \mathcal{R} \Leftrightarrow |x| \geq 0$$

- A. $\{x \in \mathcal{R}^+ \text{ and } |x| > 0\}$ and $\{|x| > 0 \text{ and } x \in \mathcal{R}\}$
 B. $\{x \in \mathcal{R} \text{ and } |x| \neq 0\}$ and $\{|x| \geq 0 \text{ and } x \notin \mathcal{R}\}$
 C. $\{x \in \mathcal{R} \text{ and } |x| \neq 0\}$ or $\{|x| \geq 0 \text{ and } x \notin \mathcal{R}\}$
 D. $\{x \in \mathcal{R}^+ \text{ and } |x| \neq 0\}$ and $\{|x| < 0 \text{ and } x \notin \mathcal{R}\}$

8. A particle of mass m moves in a straight line under the action of a resultant force F where $F = F(x)$.

Given that the velocity v is v_0 when the position is $x = x_0$, and that v is v_1 when x is x_1 , it follows that $|v_1| =$

- A. $\sqrt{\frac{2}{m} \int_{x_0}^{x_1} \sqrt{F(x)} dx} + v_0$
 B. $\sqrt{2} \int_{\sqrt{x_0}}^{\sqrt{x_1}} F(x) dx + v_0$
 C. $\sqrt{\frac{2}{m} \int_{x_0}^{x_1} F(x) dx + (v_0)^2}$
 D. $\sqrt{\frac{2}{m} \int_{x_0}^{x_1} [F(x) + (v_0)^2] dx}$

Section I continues

9. If a, b and c are real numbers such that $a^2 + b^2 + c^2 = 1$, then $ab + bc + ca$ lies in the interval

A. $\left[-\frac{1}{2}, 1\right]$

B. $\left[\frac{1}{2}, 2\right]$

C. $\left[-1, \frac{1}{2}\right]$

D. $[-1, 2]$

- 10 Grin could correctly write the value of

$$\binom{15}{0} + 3 \binom{15}{1} + 5 \binom{15}{2} + \cdots + (2n + 1) \binom{15}{n} + \cdots + 31 \binom{15}{15}.$$

What did he write down?

A. 2^{15}

B. 2^{16}

C. 2^{19}

D. 2^{31}

End of Section I

Section II

Answer each question in the appropriate writing booklet. Extra writing booklets are available.

In Questions 11 – 16, your response should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Use the Question 11 Writing Booklet.

(a) Given $z = 1 - i$, find:

(i) $\operatorname{Im} \left(\frac{1}{z} \right)$ 2

(ii) z^{10} in cartesian form 2

(iii) The two values for ω such that $\omega^2 = 3\bar{z} + i$. 3

(b) If $2 - 3i$ is a zero of the polynomial $z^3 + pz + q$ where p and q are real, find the values of p and q . 3

(c) Find 3

$$\int \frac{x^2 + 1}{x^4 + x^2 + 1} dx$$

(d) Find 2

$$\int \frac{dx}{\sqrt{3 + 4x - 4x^2}}$$

End of Question 11

Question 12 (15 marks) Use the Question 12 Writing Booklet.

- (a) Use the substitution $t = \tan \frac{\theta}{2}$ to find 3

$$\int \frac{\tan \theta}{1 + \cos \theta} d\theta$$

- (b) Consider the propositions

$$p: 2^n - 1 \text{ is prime} \quad q: n \text{ is prime,} \quad n \in \mathbb{Z}^+$$

- (i) Give an indirect proof by contrapositive of the following proposition: 3

If $2^n - 1$ is prime then n is prime

- (ii) Write down the converse of the proposition in (i) 1

- (iii) Is the converse of the proposition in (i) true? 1

If yes, give a proof.

If no, give reasons.

- (c) (i) Find real numbers A , B and C such that 3

$$\frac{9}{(2x-1)(x+1)^2} = \frac{A}{2x-1} + \frac{B}{x+1} + \frac{C}{(x+1)^2}$$

- (ii) Hence, evaluate 2

$$\int_1^2 \frac{9}{(2x-1)(x+1)^2} dx$$

- (d) The complex number z is such that $|z-1| \leq 1$ and $|z-2| = 1$. Find the maximum possible value of $|z|^2$. 2

End of Question 12

Question 13 (13 marks) Use the Question 13 Writing Booklet.

- (a) Using principle of mathematical induction, prove that $\forall n > 5, n \in \mathcal{N}$, 3

$$n! < \left(\frac{n}{2}\right)^n$$

- (b) (i) $I_n = \int x^n e^{ax} dx$, where a is a constant. 2

$$\text{Prove that } I_n = \frac{x^n e^{ax}}{a} - \frac{n}{a} I_{n-1}.$$

- (ii) Hence, find the value of 3

$$\int_0^1 x^3 e^{2x} dx$$

- (c) (i) Show that $x = r \cos(\omega t + \phi)$, where r, ω, ϕ are constants, is a solution of 1

$$\frac{d^2 x}{dt^2} = -\omega^2 x.$$

A small naval target rises and falls with SHM of a period of 10 seconds. The height of the waves from the crest to the trough is 2 metres.

At a horizontal range of 2000 m, a gun is fired so that the target would be hit provided it remains stationary in its highest position. The horizontal component of velocity is 1000 ms^{-1} .

- (ii) Show that the target would be missed by a vertical height of 0.69 m. 4

End of Question 13

Question 14 (17 marks) Use the Question 14 Writing Booklet.

- (a) A vehicle of mass m kg moves in a straight line subject to a resistance $P + Qv^2$ Newtons, where v is the speed and P and Q are constants with $Q > 0$.
- (i) Form an equation of motion for the acceleration of the vehicle. 1
- (ii) Hence, show that if $P = 0$, the distance required to slow down from the speed $\frac{3U}{2}$ to speed U is $\frac{m}{Q} \ln\left(\frac{3}{2}\right)$. 3
- (iii) Also show that if $P > 0$, the distance required to stop from speed U is given by 3

$$D = \lambda \ln(1 + kU^2)$$

Where k and λ are constants.

- (b) Let A, B, C and D represent four complex numbers z_1, z_2, z_3 and z_4 such that

$$|z_1| = |z_2| = |z_3| = |z_4| = 1 \quad \text{and} \quad z_1 + z_2 + z_3 + z_4 = 0.$$

- (i) Prove that 2
- $$|z_1 - z_2|^2 + |z_2 - z_3|^2 + |z_3 - z_4|^2 + |z_4 - z_1|^2 \geq 8$$
- (ii) α . Using (i), or otherwise, prove that 1

$$|z_1 - z_2|^2 + |z_2 - z_3|^2 + |z_3 - z_4|^2 + |z_4 - z_1|^2 =$$

$$2(|z_1|^2 + |z_2|^2 + |z_3|^2 + |z_4|^2)$$

if and only if $z_1 + z_3 = 0$ and $z_2 + z_4 = 0$.

- β . Draw an Argand diagram to illustrate this result, and explain its geometrical significance. 2

Question 14 continues

Question 14 (continued)

- (iii) Consider two complex numbers u and v defined as $|u| = |v| = 1$ and $uv \neq -1$. ω is a complex number such that $\omega = \frac{u - v}{1 + uv}$, 2

Prove that $\operatorname{Re}(\omega) = 0$.

- (iv) If $|u| < 1$, $|v| < 1$ and $uv \neq -1$, and $\xi = \frac{u - v}{1 + \bar{u}v}$, prove that 3

$$|\xi| \geq \frac{||u| - |v||}{1 - |u||v|}$$

End of Question 14

Question 15 (15 marks) Use the Question 15 Writing Booklet.

- (a) (i) Prove for every positive integer k ,

2

$$\frac{1}{k^2} < \frac{1}{k - \frac{1}{2}} - \frac{1}{k + \frac{1}{2}}$$

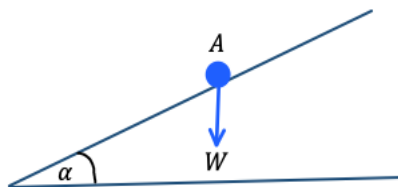
- (ii) Hence, deduce that if n is a positive integer, then

3

$$\sum_{k=1}^n \frac{1}{k^2} < 2$$

- (b) The diagram below shows a plane inclined at an angle α to the horizontal. A body of weight W can be supported on this plane independently by either the force P that is acting parallel to the incline plane, or Q that is acting parallel to the horizontal plane.

2



By drawing free body diagrams in each case, Prove that

$$\frac{1}{P^2} - \frac{1}{Q^2} = \frac{1}{W^2}$$

- (c) Find the shortest distance from the point $P(1,2,0)$ to the line with parametric equation

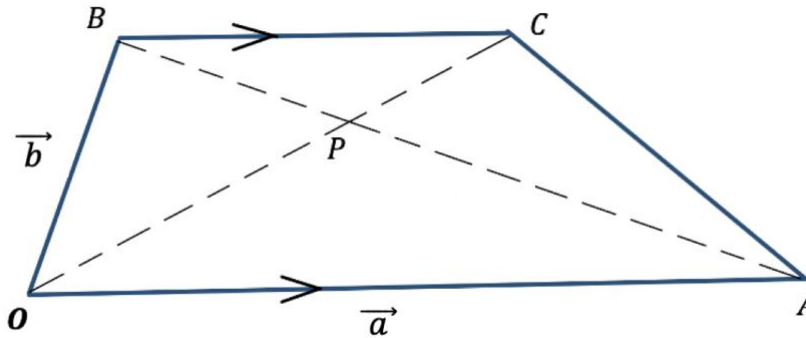
3

$$x = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}, \quad \lambda \in \mathcal{R}.$$

Question 15 continues

Question 15 (continued)

- (d) In the diagram below, $OACB$ is a trapezium, with OA and OB represented by vectors $\vec{a}$ and $\vec{b}$ respectively. The diagonals of the trapezium intersect at the point P as shown in the diagram.



- (i) Show that the vector equation of the line OC is

1

$$\vec{r} = t(\vec{b} + \alpha \vec{a}), \quad t, \alpha \in \mathcal{R}.$$

- (ii) Prove by vector methods that the point of intersection of the diagonals of a trapezium lies on the line passing through the midpoints of the parallel sides.

4

End of Question 15

Question 16 (15 marks) Use the Question 16 Writing Booklet.

- (a) Prove that if a real number c satisfies a polynomial equation of the form 4

$$r_3x^3 + r_2x^2 + r_1x + r_0 = 0,$$

where, r_0, r_1, r_2 and r_3 are rational numbers, then c satisfies an equation of the form

$$n_3x^3 + n_2x^2 + n_1x + n_0 = 0,$$

where n_0, n_1, n_2 and n_3 are integers.

- (b) Consider the compound function 2

$$f(x(\theta)) = h + x(\theta)\tan\theta - \frac{g}{2v^2}(x(\theta))^2 \sec^2 \theta,$$

where g, h and v are constants and $0 < \theta < \frac{\pi}{2}$.

Using implicit differentiation or otherwise, Prove that the value of θ that maximises $x(\theta)$ can be given by

$$x_m = \frac{v^2}{g} \cot \theta_m$$

NOTE: θ_m is the angle that gives the maximum value x_m

- (c) A projectile of unit mass launched from a tower of height h metres with velocity v m/s at an angle of θ .

The equations of motion of the particle are as shown below:

$$\vec{r} = \begin{pmatrix} v \cos \theta t \\ h + v \sin \theta t - \frac{1}{2}gt^2 \end{pmatrix}$$

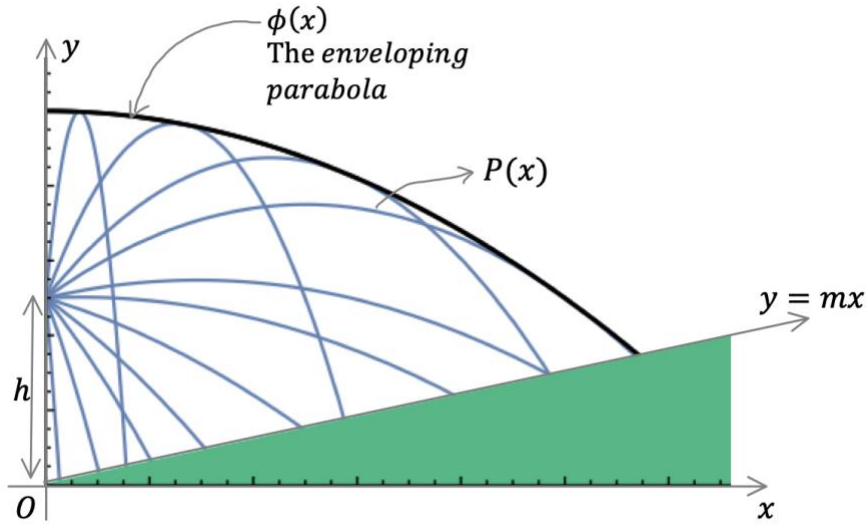
DO NOT PROVE THIS

- (i) Show that, for a given launch velocity v and angle θ , the Cartesian equation of the trajectory of the particle is given by 1

$$y = P(x) = h + x\tan\theta - \frac{gx^2}{2v^2} \sec^2 \theta$$

Question 16 continues

The diagram below shows the trajectory of the projectile launched at velocity v m/s for varying values of θ . The function $\phi(x)$ is the enveloping parabola that encloses all possible paths, including the highest and the longest.



- (ii) For any **given value of x** within the range of the projectile and launch velocity v , show that the equation of the enveloping parabola is

3

$$y = h + \frac{v^2}{2g} - \frac{gx^2}{2v^2}$$

(HINT: you may set $u = \tan \theta$)

- (iii) The projectile impacts on an inclined plane with equation $y = mx$ at the point $x = c$ as shown in the diagram above. Show that the x - coordinate of the point of intersection of the projectile and the inclined plane is given by

2

$$c = -\frac{mv^2}{g} \pm \frac{v^2}{g} \sqrt{m^2 + \frac{2gh}{v^2} + 1}$$

- (iv) Hence, show that the optimal angle θ_m that gives the largest range is given by

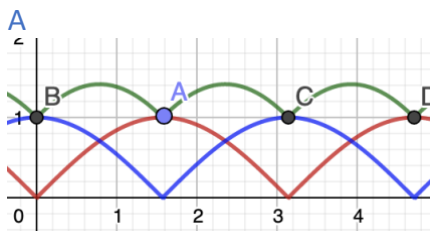
3

$$\theta_m = \cot^{-1} \left(-m + \sqrt{m^2 + \frac{2gh}{v^2} + 1} \right)$$

(You may refer to the result in (b).)

End of Examination

2021 Ext 2 Trial Marking scheme

No:	Sample Answer	Marking scheme	Feedback
1	A $\operatorname{Arg} z_1 = \frac{\pi}{4} \quad \text{1st quadrant}$ $\operatorname{Arg} z_2 = \frac{\pi}{6} \quad \text{1st quadrant}$ $\operatorname{Arg} \frac{z_1}{z_2} = \frac{\pi}{4} - \frac{\pi}{6} = \frac{\pi}{12} \quad \text{1st quadrant}$ Also, $\operatorname{Arg} z_2 < \operatorname{Arg} z_1$		
2	B $\frac{F \cdot v}{v \cdot v} v = \frac{2 + 8 + 16}{4 + 16 + 16} \begin{pmatrix} 2 \\ 4 \\ -4 \end{pmatrix} = \frac{13}{9} \begin{pmatrix} 1 \\ 2 \\ -2 \end{pmatrix}$		
3	C $\left(\frac{1}{c}\right)^2 + \left(\frac{1}{c}\right)^2 + \left(\frac{1}{c}\right)^2 = 1$ $\frac{3}{c^2} = 1 \rightarrow c^2 = 3 \rightarrow c = \pm\sqrt{3}$	Sum of squares of direction cosines is 1.	
4	A 		
5	D $z^n = \sqrt{2} e^{i(\frac{\pi}{4} + 2\pi k)}, \quad k \in \mathbb{Z}$ $z = 2^{\frac{1}{2n}} e^{i(\frac{\pi}{4n} + \frac{2\pi k}{n})}, \quad k \in \mathbb{Z}$		
6	B $F(t) = \int_0^1 f(t-y)g(y)dy$		

	$= \int_0^t f e^{t-y} y dy$ Let $t - y = p$ $-dy = dp$; Also, $y = 0, p = t; y = t, p = 0$ $\therefore F(t) = - \int_1^0 (t - p) e^p dp$ $= \int_0^1 t e^p dp - \int_0^1 p e^p dp$ $= t[e^p]_0^1 - [pe^p]_0^1 + [e^p]_0^1$ $= [t(e^1 - 1) - te^1 + (e^1 - 1)]$ $F(t) = e^t - (t + 1)$		
7	C Here $p: x \in \mathcal{R}$ and $q: x \geq 0$ $p \Leftrightarrow q$ is a biconditional statement. $\neg(p \Leftrightarrow q) = \neg(p \Rightarrow q \text{ and } q \Rightarrow p)$ $= \neg(p \Rightarrow q) \text{ or } \neg(q \Rightarrow p)$ $= (p \text{ and } \neg q) \text{ or } (q \text{ and } \neg p)$ Solution: $\{x \in \mathcal{R} \text{ and } x \neq 0\} \text{ or } \{ x \geq 0 \text{ and } x \notin \mathcal{R}\}$		

8	C $m\ddot{x} = F(x)$ $\ddot{x} = \frac{F(x)}{m}$ $v \frac{dv}{dx} = \frac{1}{m} F(x)$ $\int_{v_0}^{v_1} v \, dv = \frac{1}{m} \int_{x_0}^{x_1} F(x) \, dx$ $\frac{1}{2} ((v_1)^2 - (v_0)^2) = \frac{1}{m} \int_{x_0}^{x_1} F(x) \, dx$ $(v_1)^2 = \frac{2}{m} \int_{x_0}^{x_1} F(x) \, dx + (v_0)^2$ $v_1 = \sqrt{\frac{2}{m} \int_{x_0}^{x_1} F(x) \, dx + (v_0)^2}$		
9.	A Let $ab + bc + ca = k$ Now, $(a + b + c)^2 = a^2 + b^2 + c^2 + 2k$ $(a + b + c)^2 \geq 0$ perfect square $1 + 2k \geq 0 \Rightarrow k \geq -\frac{1}{2}$ Consider $(a - b)^2 + (b - c)^2 + (c - a)^2 \geq 0$ $2[a^2 + b^2 + c^2 - ab - bc - ca] \geq 0$ $1 - k \geq 0 \rightarrow k \leq 1$ $\left[-\frac{1}{2}, 1\right]$		
10	C $I = \binom{15}{0} + 3 \binom{15}{1} + 5 \binom{15}{2} + \dots + (2n + 1) \binom{15}{n} + \dots + 31 \binom{15}{15}.$ $J = 31 \binom{15}{15} + 29 \binom{15}{14} + 27 \binom{15}{13} + \dots + (31 - 2n) \binom{15}{15 - n} + \dots + \binom{15}{0}.$ J is the same as I, but the terms are arranged in the reverse order as $\binom{n}{r} = \binom{n}{n - r}.$ Hence, $\binom{15}{0} = \binom{15}{15}; \quad \binom{15}{1} = \binom{15}{14} \dots \dots$		

	$I + J = 32 \left[\binom{15}{0} + \binom{15}{1} + \binom{15}{2} + \cdots + \binom{15}{n} + \cdots + \binom{15}{15} \right].$ $2I = 32 \times 2^{15} = 2^{18}$		

Question 11

a)(i)	$\frac{1}{z} = \frac{1}{1-i} = \frac{1+i}{2}$ $\operatorname{Im} \left(\frac{1}{z} \right) = \frac{1}{2}$	1 mark: multiplies by the conjugate 1 mark: gives the imaginary component	
a(ii)	$z^{10} = \left(\sqrt{2} e^{-\frac{i\pi}{4}} \right)^{10}$ $= 32 e^{-\frac{10\pi}{4}} = 32 e^{-\frac{\pi}{2}} = -32i$	1 mark: expresses z^{10} in component form of exponential form. 1 mark: correctly evaluates and gives the answer in Cartesian form.	
a(iii)	$\omega^2 = 3\bar{z} + i = 3(1+i) + i$ $\omega^2 = 3 + 4i = (a+ib)^2$ $a^2 - b^2 = 3$ $a^2 + b^2 = \sqrt{(a^2 - b^2)^2 + (2ab)^2}$ $a^2 + b^2 = \sqrt{3^2 + 4^2} = 5$ Hence, $2a^2 = 8 \rightarrow a = \pm 2$ Hence, $2b^2 = 2 \rightarrow b = \pm 1$ $\text{Thus } \omega^2 = \sqrt{(2+i)^2} = \pm(2+i)$	3 marks: Correct answer from correct working 2 marks: Equates the real and imaginary parts and writes the relationship attempting to find a and b, writes the squareroot from their a and b 1 mark: Equates the real and imaginary parts, but equations or method incorrect	
b)	$z^3 + pz + q = 0$ $2 - 3i$ is a root	1 mark: substitutes into the polynomial	

	$(2 - 3i)^3 + (2 - 3i)p + q = 0 \quad \text{1 mark}$ $8 - 36i - 54 + 27i + (2 - 3i)p + q = 0$ $-46 - 9i + 2p + q - 3ip = 0$ $2p + q = 46 \quad 3p = -9; \quad \text{1 mark}$ $p = -3$ $\therefore q = 46 - 2 \times -3 = 52 \quad \text{1 mark}$ Or Let the roots be $\alpha, \bar{\alpha}, \beta$ Sum of roots is zero. $(2 - 3i) + (2 + 3i) + \beta = 0$ $\therefore \beta = -4$ Product of roots = $(2 - 3i)(2 + 3i)(-4) = -q$ $\therefore q = 52$ $p = (2 - 3i)(-4) + (2 + 3i)(-4) + (2 - 3i)(2 + 3i)$ $= -3$	1 mark: equates the real and imaginary parts 1 mark: solves for p and q correctly	
c)	$\int \frac{x^2 + 1}{x^4 + x^2 + 1} dx$ $\int \frac{1 + \frac{1}{x^2}}{x^2 + 1 + \frac{1}{x^2}} dx$ $\text{let } u = x - \frac{1}{x}$ $du = 1 + \frac{1}{x^2} dx \quad (\text{give 1 mark here})$ $\int \frac{1 + \frac{1}{x^2}}{x^2 + 1 + \frac{1}{x^2}} dx = \int \frac{1 + \frac{1}{x^2}}{\left(x - \frac{1}{x}\right)^2 + 3} dx \quad \text{2 mark}$	2 mark: reorganises the integrand, chooses the correct substitution, expresses dx and the integrand into an integrable form 1 mark: minor error or significant progress with relevant working	

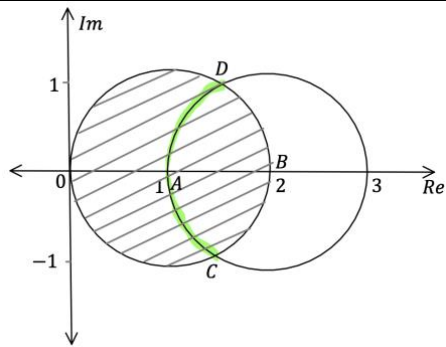
	$\int \frac{du}{u^2 + 3} = \frac{1}{\sqrt{3}} \tan^{-1} \frac{u}{\sqrt{3}} + C$ $\frac{1}{\sqrt{3}} \tan^{-1} \frac{x - \frac{1}{x}}{\sqrt{3}} + C$ $\frac{1}{\sqrt{3}} \tan^{-1} \frac{x^2 - 1}{\sqrt{3}x} + C$	1 mark: correctly integrates and expresses the solution in terms of x	
d)	$\int \frac{dx}{\sqrt{3 + 4x - 4x^2}} =$ Since, $3 + 4x - 4x^2 = 4\left(\frac{3}{4} + x - x^2\right)$ $4\left(1 - \left(x - \frac{1}{2}\right)^2\right)$ $\int \frac{dx}{\sqrt{4\left(1 - \left(x - \frac{1}{2}\right)^2\right)}} = \frac{1}{2} \int \frac{dx}{\sqrt{1 - \left(x - \frac{1}{2}\right)^2}}$ Let $u = x - \frac{1}{2}$ $du = dx$ $\frac{1}{2} \int \frac{du}{\sqrt{1 - u^2}} = \frac{1}{2} \sin^{-1} u = \frac{1}{2} \sin^{-1} \left(x - \frac{1}{2}\right) + C$ $\frac{1}{2} \sin^{-1} \left(\frac{2x - 1}{2}\right) + C$	1 mark: completes the square and expresses the integrand in the $\frac{dx}{\sqrt{1 - \left(x - \frac{1}{2}\right)^2}}$ form 1 mark: correctly integrates and expresses the answer in terms of x.	

Question 12

a)	Let $t = \tan \frac{\theta}{2}$, then $\tan \theta = \frac{2t}{1-t^2}, \cos \theta = \frac{1-t^2}{1+t^2} \text{ and } \frac{d\theta}{dt} = \frac{2}{1+t^2}$ $\int \frac{\tan \theta}{1+\cos \theta} d\theta = \int \frac{2t}{1-t^2} \div \left(1 + \frac{1-t^2}{1+t^2}\right) \times \frac{2}{1+t^2} dt$ $= \int \frac{2t}{1-t^2} \times \left(\frac{1+t^2}{2}\right) \times \frac{2}{1+t^2} dt$ $= \int \frac{2t}{1-t^2} dt$ $= -\ln 1-t^2 + C$ $= -\ln\left 1-\tan^2 \frac{\theta}{2}\right + C$	3 mark: correct solution showing all working 2 mark: correctly converts the integrand, but error in simplification or integration 1 mark: correct conversion of integrand Or correct integration of the integrand in t.	
b)(i)	$p: 2^n - 1$ is prime $q: n$ is prime We need to prove $p \Rightarrow q$ We have $p \Rightarrow q \equiv \neg q \Rightarrow \neg p$ We'll prove the contrapositive statement $\neg q \Rightarrow \neg p$ Proof: $\neg q: n$ is composite. 1 mark Then $\exists a, b \in \mathbb{Z}^+: n = ab, a \neq 1, b \neq 1$. Hence, $2^{ab} - 1 = (2^a)^b - 1$ $= (2^a - 1)(2^{a(b-1)} + 2^{a(b-2)} + 2^{a(b-3)} + \dots + 1)$ Where $2^a - 1 \neq 1$, since $a \neq 1$ and	* writes the contrapositive statement correctly *expresses $2^a - 1$ with n as a composite number *clearly states, $b \in \mathbb{Z}^+: n = ab, a \neq 1, b \neq 1$. *expresses $2^n - 1$ in factored form * Clearly explains $2^a - 1 \neq 1$ *States $2^a - 1 \neq 2^{ab} - 1$ since $b \neq 1$	

	$2^a - 1 \neq 2^{ab} - 1$ since $b \neq 1$ So, $2^n - 1$ is composite, $\neg p$ is true. We have proved $\neg q \Rightarrow \neg p$ Hence, $p \Rightarrow q$ is true.	*states the contrapositive statement and hence the given proposition is true. 3 marks: All 7 components are clearly demonstrates 2 mark: at least five 1 mark: at least 3	
b(ii)	The converse of $p \Rightarrow q$ is $q \Rightarrow p$. If n is prime, then $2^n - 1$ is prime.	1 mark:	
b(iii)	The converse is false. Let $n = 11$, $2^{11} - 1 = 23 \times 89$ is composite, thus disproving $q \Rightarrow p$	1 mark: reasons correctly	
c(i)	$\frac{9}{(2x-1)(x+1)^2} = \frac{A}{2x-1} + \frac{B}{x+1} + \frac{C}{(x+1)^2}$ $9 = A(x+1)^2 + B(2x-1)(x+1) + C(2x-1)$ $x = -1, \quad 9 = -3C \Rightarrow C = -3$ $x = \frac{1}{2}, \quad 9 = \frac{9}{4}A \Rightarrow A = 4$ Compare constants, $9 = A - B - C \Rightarrow B = -2$ $\frac{9}{(2x-1)(x+1)^2} = \frac{4}{2x-1} - \frac{2}{x+1} - \frac{3}{(x+1)^2}$ Or $x = \frac{1}{2}, \quad A = \frac{9}{\left(\frac{3}{2}\right)^2} = 4$ $x = -1, \quad C = \frac{9}{2(-1)-1} = -3$	3 marks: All pronumerals are calculated correctly showing all working 2 marks: two of the pronumerals are calculated correctly showing all working 1 mark: A valid method is used to evaluate the constants.	

	Let $x = 0$, then $\frac{9}{-1 \times -1} = \frac{4}{-1} + B - 3$ $\therefore B = -2$		
c)(ii)	$\int_1^2 \frac{9}{(2x-1)(x+1)^2} dx =$ $\int_1^2 \frac{4}{2x-1} - \frac{2}{x+1} - \frac{3}{(x+1)^2} dx$ $\left[2 \ln 2x-1 - 2 \ln x+1 - \frac{3(x+1)^{-1}}{-1} \right]_1^2$ $= \left[2 \ln \left \frac{2x-1}{x+1} \right + \frac{3}{x+1} \right]_1^2$ $= 2 \ln \left \frac{1}{\frac{1}{2}} \right + 3 \left(\frac{1}{3} - \frac{1}{2} \right)$ $= 2 \ln 2 - \frac{1}{2} \quad \text{or} \quad \ln 4 - \frac{1}{2}$	1 mark: correctly integrates 1 mark: correct evaluation (must give the answer in the simplest form)	
d)	$z-1 \leq 1$ and $z-2 = 1$. $z-1 \leq 1$ is the circle and the interior of circle with centre (1, 0) and radius 1. $z-2 = 1$ circle with centre (2,0) and radius 1. Both these conditions are satisfied by arc DAC.	2 marks: Correct answer from correct working. 1 mark: Minor error or explains arc DAC satisfies the two conditions	



$$OA \leq |z| \leq OC. \text{ or } OD$$

As $\angle OCB = \frac{\pi}{2}$,

$$\Rightarrow OC^2 = OB^2 - BC^2 = 4 - 1 = 3$$

Hence, $|z|^2 \leq 3$

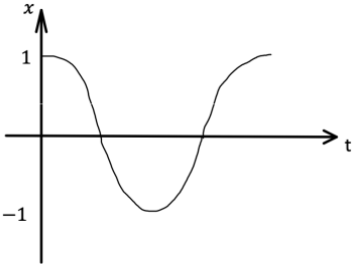
1 mark: uses the triangular inequality to obtain the inequation

1 mark: solves the equation and gives the greatest value

Question 13

a)	$p(n): n! < \left(\frac{n}{2}\right)^n, n > 5. n \in \mathcal{N}$ Step 1: $n = 6$ $6! < \left(\frac{6}{2}\right)^6 \Rightarrow 6! < 3^6 \Rightarrow 720 < 729$ True $\therefore p(6)$ is true. 1 mark Step 2 Suppose $p(k)$ is true. $p(k): k! < \left(\frac{k}{2}\right)^k, \quad k > 5 \quad (1)$ Proving $p(k+1)$ is true. Multiplying (1) by $(k+1)$ $p(k+1): k! (k+1) < \left(\frac{k}{2}\right)^k (k+1)$ $(k+1)! < \frac{k^k}{2^k} (k+1)$ $(k+1)! < \frac{k^k(k+1) \times 2}{2^k \times 2} \quad (2) \quad 1 \text{ mark}$ Consider $\left(1 + \frac{1}{k}\right)^k$ $\left(1 + \frac{1}{k}\right)^k = 1 + k \left(\frac{1}{k}\right) + \binom{k}{2} \left(\frac{1}{k}\right)^2 + \dots$ Hence, $\left(1 + \frac{1}{k}\right)^k > 2$ Hence, from (2) $(k+1)! < \frac{k^k(k+1) \times 2}{2^k \times 2} < \frac{k^k(k+1)}{2^{k+1}} \left(1 + \frac{1}{k}\right)^k$	1 mark: Proves the basic result 1 mark: writes the $p(k)$ and $p(k+1)$ statements and demonstrates the basic manipulation 2 mark: proves $\left(1 + \frac{1}{k}\right)^k > 2$ and proves $p(k+1)$ is true and gives a concluding statement 1 mark: attempts and reasonable manipulation to prove the inequality and gives a concluding statement	
----	--	---	--

	$\therefore (k+1)! < \frac{k^k(k+1)}{2^{k+1}} \times \frac{(k+1)^k}{k^k}$ $\therefore (k+1)! < \frac{(k+1)^{k+1}}{2^{k+1}}$ Hence, $p(k+1)$ is true. Thus $p(k)$ is true $\Rightarrow p(k+1)$ is true for $\forall k > 5, k \in \mathcal{N}$ $\therefore$ by principle of mathematical induction, $p(n)$ is true for $\forall n > 5, n \in \mathcal{N}$		
(b)(i)	$I_n = \int x^n e^{ax} dx$ Let $u = x^n, u' = nx^{n-1}$ $v' = e^{ax} \quad v = \frac{e^{ax}}{a}$ $I_n = \frac{x^n}{a} e^{ax} - \frac{n}{a} \int x^{n-1} e^{ax} dx$ $I_n = \frac{x^n}{a} e^{ax} - \frac{n}{a} I_{n-1} \text{ as required.}$	1 mark: gives the correct expressions for u, u', v, v' or significant progress towards solution 1 mark: gives correct steps to conclusion	
b(ii)	$I_3 = \int_0^1 x^3 e^{2x} dx$ $I_3 = \left[\frac{x^3}{2} e^{2x} \right]_0^1 - \frac{3}{2} I_2 = \frac{e^2}{2} - \frac{3}{2} I_2$ $I_2 = \left[\frac{x^2}{2} e^{2x} \right]_0^1 - \frac{2}{2} I_1 = \frac{e^2}{2} - I_1$ $I_1 = \left[\frac{x}{2} e^{2x} \right]_0^1 - \frac{1}{2} I_0 = \frac{e^2}{2} - \frac{1}{2} I_0$ $I_0 = \int_0^1 e^{2x} dx = \left[\frac{1}{2} e^{2x} \right]_0^1 = \frac{e^2}{2} - \frac{1}{2}$	1 mark: some progress towards solution. Eg. One correct step say from I_3 to I_2. 1 mark: gives correct expression for I_0 1 mark: gives correct solution	

	$I_1 = \frac{e^2}{2} - \frac{1}{2} \left(\frac{e^2}{2} - \frac{1}{2} \right) = \frac{e^2}{2} - \frac{e^2}{4} + \frac{1}{4} = \frac{e^2}{4} + \frac{1}{4}$ $I_2 = \frac{e^2}{2} - \frac{1}{2} \left(\frac{e^2}{2} - \frac{1}{2} \right) = \frac{e^2}{2} - \left(\frac{e^2}{4} + \frac{1}{4} \right) = \frac{e^2}{4} - \frac{1}{4}$ $I_3 = \frac{e^2}{2} - \frac{3}{2} I_2 = \frac{e^2}{2} - \frac{3}{2} \left(\frac{e^2}{4} - \frac{1}{4} \right) = \frac{e^2}{8} + \frac{3}{8}$		
(c)(i)	$x = r \cos(\omega t + \phi)$ $\dot{x} = -r\omega \sin(\omega t + \phi)$ $\ddot{x} = -r\omega^2 \sin(\omega t + \phi)$ $\ddot{x} = -\omega^2 x$ ie. $r \cos(\omega t + \phi)$ satisfies the equation $\ddot{x} = -\omega^2 x$.	1 mark: proves the result with all lines of working	
C(ii)	Crest to trough is 2m $\therefore$ amplitude $a = 1 \text{ m}$ We are considering the naval vessel while it is on the crest and the bullet will hit it provided it stayed stationary.  To travel a distance of 2000 m, the bullet takes $\frac{2000 \text{ m}}{1000 \text{ m/s}} = 2 \text{ sec}$ We will calculate the vertical height (x) of the vessel after 2 seconds. Period $T = \frac{2\pi}{n}$	1 mark: calculates the time taken for a distance of 2000m. $t = 2$ 2 marks: calculates a, n and α and writes the equation of motion of the boat. 1 mark: minor error in calculating a, n and α and writes the equation of motion of the boat.	

$$10 = \frac{2\pi}{n} \Rightarrow n = \frac{2\pi}{10} = \frac{\pi}{5}$$

$$x = a \cos(nt + \alpha)$$

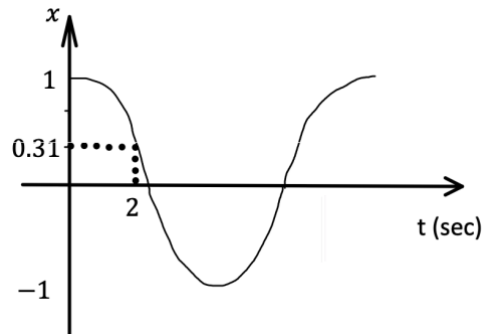
$$x = 1 \cos(nt + \alpha)$$

At $t = 0, x = 1$ (at the crest)

$$\therefore \cos \alpha = 1 \quad \text{ie. } \alpha = 0$$

Hence, $x = 1 \cos\left(\frac{\pi}{5}t\right)$

When $t = 2$, $x = \cos \frac{2\pi}{5} = 0.309 \dots m$



ie. the difference in height of the bullet and the target is

$$1 \text{ m} - 0.309 \dots m$$

$$= 0.6909 \dots$$

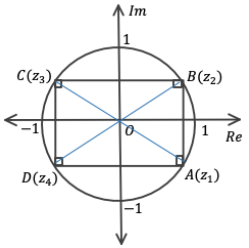
$$\approx 0.69 \text{ m}$$

1 mark: calculates the displacement of the boat at $t = 2$ and calculates the difference in displacements of the bullet and the boat.

Question 14

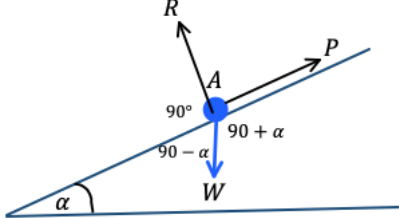
14(a) (i)	 $m\ddot{x} = -(P + Qv^2)$ $\ddot{x} = -\frac{1}{m}(P + Qv^2)$	1 mark: correct free body diagram, equation of motion and derives for $\ddot{x}$	
(ii)	$\ddot{x} = -\frac{1}{m}(P + Qv^2)$ If $P = 0$, $\ddot{x} = -\frac{Q}{m}v^2$ $\ddot{x} = -\frac{Q}{m}v^2$ $\ddot{x} = -\frac{Q}{m}v^2$ $v \frac{dv}{dx} = -\frac{Q}{m}v^2$ Hence, $\frac{dx}{dv} = -\frac{m}{Qv}$ when $x = 0, v = \frac{3U}{2}$ and when $x = D, v = U$ Hence, $\int_0^D dx = -\frac{m}{Q} \int_{\frac{3U}{2}}^U \frac{1}{v} dv$ $D = -\frac{m}{Q} \ln \left \frac{U}{\frac{3U}{2}} \right $ $D = -\frac{m}{Q} \ln \left \frac{2}{3} \right = \frac{m}{Q} \ln \left \frac{3}{2} \right $	1 mark: expresses $\ddot{x} = v \frac{dv}{dx}$ and writes the equation 1 mark: separates the differential equation into variable separable form 1 mark: Integrates, evaluates and simplifies	

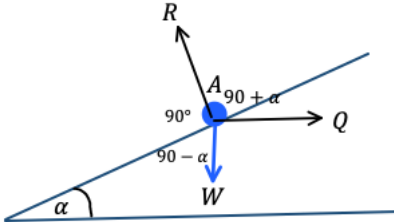
(iii)	If $P > 0$, then $\frac{dv}{dx} = -\left(\frac{P + Qv^2}{mv}\right)$ Hence, $\frac{dx}{dv} = -\frac{mv}{P + Qv^2}$ Set the distance at which $V = U$ as $x = 0$, and $V = 0$, when, say $x = D$ $\frac{dx}{dv} = -\frac{mv}{P + Qv^2} \quad 1 \text{ mark}$ $\int_0^D dx = \int_U^0 -\frac{mv}{P + Qv^2} dv$ $D = -\frac{m}{2Q} \int_U^0 \frac{2Qv}{P + Qv^2} dv$ $D = -\frac{m}{2Q} [\ln P + Qv^2]_U^0 = -\frac{m}{2Q} \ln\left \frac{P}{P + QU^2}\right \quad 1 \text{ mark}$ $= \frac{m}{2Q} \ln\left \frac{P + QU^2}{P}\right $ $= \frac{m}{2Q} \ln\left 1 + \frac{Q}{P}U^2\right \equiv \lambda \ln(1 + kU^2)$ Hence, $\lambda = \frac{m}{2Q}$ and $k = \frac{Q}{P}$ 1 mark	1 mark: gives the correct differential equation 1 mark: Integrates the differential equation correctly 1 mark: correctly simplifies to give the values of λ and k.	
14 (b)(i)	We know, $ z_1 - z_2 ^2 = (z_1 - z_2)(\overline{z_1 - z_2})$ $= (z_1 - z_2)(\bar{z}_1 - \bar{z}_2)$ $= z_1\bar{z}_1 - z_1\bar{z}_2 - z_2\bar{z}_1 + z_2\bar{z}_2 = z_1 ^2 + z_2 ^2 - z_1\bar{z}_2 - z_2\bar{z}_1$ Let $E = z_1 - z_2 ^2 + z_2 - z_3 ^2 + z_3 - z_4 ^2 + z_4 - z_1 ^2$	2 marks: Correct proof and reasoning 1 mark: expresses $z ^2$ as $z\bar{z}$ and uses	

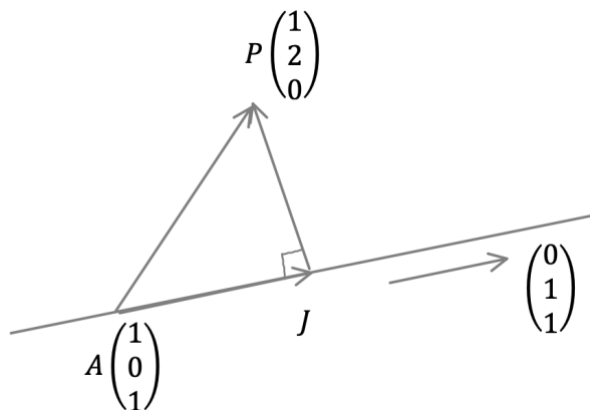
	$E = 2(z_1 ^2 + z_2 ^2 + z_3 ^2 + z_4 ^2) - \bar{z}_1(z_2 + z_4) - \bar{z}_2(z_1 + z_3) - \bar{z}_3(z_2 + z_4) - \bar{z}_4(z_1 + z_3)$ $= 2(z_1 ^2 + z_2 ^2 + z_3 ^2 + z_4 ^2) - (z_2 + z_4)(\bar{z}_1 + \bar{z}_3) - (z_1 + z_3)(\bar{z}_2 + \bar{z}_4)$ However, $z_2 + z_4 = -(z_1 + z_3)$ as $z_1 + z_2 + z_3 + z_4 = 0$. $= 2(z_1 ^2 + z_2 ^2 + z_3 ^2 + z_4 ^2) + (z_1 + z_3)(\bar{z}_1 + \bar{z}_3) + (z_2 + z_4)(\bar{z}_2 + \bar{z}_4)$ Hence, $ z_1 - z_2 ^2 + z_2 - z_3 ^2 + z_3 - z_4 ^2 + z_4 - z_1 ^2 =$ $= 2(z_1 ^2 + z_2 ^2 + z_3 ^2 + z_4 ^2) + z_1 + z_3 ^2 + z_2 + z_4 ^2$ (statement 1) $= 2 \times 4 + z_1 + z_3 ^2 + z_2 + z_4 ^2$ Hence, $ z_1 - z_2 ^2 + z_2 - z_3 ^2 + z_3 - z_4 ^2 + z_4 - z_1 ^2 \geq 8$	$ z_1 = z_2 = z_3 = z_4 = 1$	
b(ii) α	From (i), using statement 1, $E - 2(z_1 ^2 + z_2 ^2 + z_3 ^2 + z_4 ^2)$ $= z_1 + z_3 ^2 + z_2 + z_4 ^2 = 0$ $\Leftrightarrow z_1 + z_3 = 0 \quad \text{and} \quad z_2 + z_4 = 0.$	1 mark: proves the result	
β	 The sum of squares of the side lengths of a rectangle with vertices of the form $z = \pm a \pm bi$ equals 8	1 mark: for correct diagram 1 mark: for the geometrical reasoning	
(iii)	$\omega + \bar{\omega} = 2 \operatorname{Re}(\omega)$ $\omega + \bar{\omega} = \frac{u-v}{1+uv} + \overline{\left(\frac{u-v}{1+uv}\right)}$ $= \frac{u-v}{1+uv} + \frac{\bar{u}-\bar{v}}{1+\bar{u}\bar{v}}$	2 mark: correct solution from correct working 1 mark: explains $\omega + \bar{\omega} = 2 \operatorname{Re}(\omega)$ and attempts to simplify the expression	

	We expand and simplify: $= \frac{u-v}{1+uv} + \frac{\bar{u}-\bar{v}}{1+\bar{u}\bar{v}}$ $= \frac{u-v+u\bar{u}\bar{v}-\bar{u}v\bar{v}+\bar{u}-\bar{v}+u\bar{u}v-uv\bar{v}}{(1+uv)(1-\bar{u}\bar{v})}$ $u\bar{u} = 1$ and $v\bar{v} = 1$ $\therefore \frac{u-v+u\bar{u}\bar{v}-\bar{u}v\bar{v}+\bar{u}-\bar{v}+u\bar{u}v-uv\bar{v}}{(1+uv)(1-\bar{u}\bar{v})} = 0$ $= \frac{u-v}{1+uv} + \frac{v-u}{1+uv} = 0$ ie. $2 \operatorname{Re}(\omega) = 0$ $\therefore \operatorname{Re}(\omega) = 0$		
(iv)	Let $u = r_1 e^{i\theta}$ and $v = r_2 e^{i\alpha}$, where $r_1 < 1, r_2 < 1$ $1 - \bar{u}v = 1 - r_1 r_2 e^{i(\alpha-\theta)}$ $ 1 - \bar{u}v ^2 = (1 - r_1 r_2 \cos(\alpha - \theta))^2 + (r_1 r_2)^2 \sin^2(\alpha - \theta)$ $= 1 - 2r_1 r_2 \cos(\alpha - \theta) + r_1^2 r_2^2$ $\leq (1 - r_1 r_2)^2 \text{ as } -1 \leq \cos(\alpha - \theta) \leq 1$ $= (1 - u v)^2$ That is $ 1 - \bar{u}v ^2 \leq (1 - u v)^2$ Hence, $\frac{1}{1- u v } \leq \frac{1}{ 1-\bar{u}v } \quad (1) \quad 1 \text{ mark (result 1)}$ Also, using triangular inequalities, $ u - v \leq u - v \quad (2) \quad 1 \text{ mark and conclusion}$ From (1) and (2), $\frac{ u - v }{1 - u v } \leq \left \frac{u - v}{1 + \bar{u}v} \right $	1 mark: expresses u and v correctly and writes an expression for $1 - \bar{u}v ^2$ 1 mark: proves the inequality result 1 1 mark: Uses triangular inequality and proves the result	

Question 15

a)(i)	$\frac{1}{k - \frac{1}{2}} - \frac{1}{k + \frac{1}{2}} = \frac{k + \frac{1}{2} - \left(k - \frac{1}{2}\right)}{k^2 - \frac{1}{4}}$ $= \frac{1}{k^2 - \frac{1}{4}} > \frac{1}{k^2}$ Hence, $\frac{1}{k^2} < \frac{1}{k - \frac{1}{2}} - \frac{1}{k + \frac{1}{2}}$	2 marks: correct proof 1 mark: algebraic manipulation and some relevant logic of inequality demonstrated	
a)(ii)	From (i), $\sum_{k=1}^n \frac{1}{k^2} < \sum_{k=1}^n \left(\frac{1}{k - \frac{1}{2}} - \frac{1}{k + \frac{1}{2}} \right) \quad 1 \text{ mark}$ $< \left(\frac{1}{\frac{1}{2}} - \frac{1}{\frac{3}{2}} \right) + \left(\frac{1}{\frac{3}{2}} - \frac{1}{\frac{5}{2}} \right) + \dots + \left(\frac{1}{n - \frac{1}{2}} - \frac{1}{n + \frac{1}{2}} \right)$ $< \frac{1}{\frac{1}{2}} - \frac{1}{n + \frac{1}{2}} \quad 1 \text{ mark}$ $< \frac{2}{1} - \frac{1}{n + \frac{1}{2}}$ $< 2 \quad \text{as } \frac{1}{n + \frac{1}{2}} > 0 \forall n \in \mathbb{Z}^+$ 1 mark	1 mark: using result (i), writes the inequality of the sum of the series 1 mark: simplifies the series Proves the result with reasoning and appropriate mathematical notations	
b)		2 marks: correct answer from correct working	

	Using Lami's theorem, $\frac{P}{\sin (180 - \alpha)} = \frac{W}{\sin 90} = \frac{R}{\sin (90 + \alpha)}$ Hence, $P = W \sin \alpha$  $\frac{Q}{\sin (180 - \alpha)} = \frac{W}{\sin (90 + \alpha)} = \frac{R}{\sin 90}$ Hence, $Q = W \tan \alpha$ $\frac{1}{P^2} - \frac{1}{Q^2} = \frac{1}{W^2} (\operatorname{cosec}^2 \alpha - \cot^2 \alpha)$ $\frac{1}{P^2} - \frac{1}{Q^2} = \frac{1}{W^2}$	1 mark: Draws the free body diagram, writes the force equations	
c)	Let A be the point $\begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$ and $P \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix}$ $\overrightarrow{AP} = \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix} - \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 2 \\ -1 \end{pmatrix}$ $\underset{\sim}{v} = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$	1 mark: calculates $\overrightarrow{AP}$, $\overrightarrow{AP} \cdot v$ correctly 1 mark: correctly calculates the $\operatorname{Proj}_v \overrightarrow{AP}$ 1 mark: calculates the shortest distance	



$$Proj_v \overrightarrow{AP} = \frac{\overrightarrow{AP} \cdot v}{v \cdot v} v$$

$$\overrightarrow{AP} \cdot v = \begin{pmatrix} 0 \\ 2 \\ -1 \end{pmatrix} \cdot \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} = 0 + 2 - 1 = 1 \quad \text{1 mark}$$

$$v \cdot v = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} = 0 + 1 + 1 = 2$$

$$Proj_v \overrightarrow{AP} = \frac{\overrightarrow{AP} \cdot v}{v \cdot v} v = \frac{1}{2} \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$$

The shortest distance =

$$= |\overrightarrow{AP} - Proj_v \overrightarrow{AP}| = \left| \begin{pmatrix} 0 \\ 2 \\ -1 \end{pmatrix} - \frac{1}{2} \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} \right|$$

$$= \frac{3}{2} \left| \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix} \right| = \frac{3}{\sqrt{2}}$$

Method 2

$$A \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}, \quad J \begin{pmatrix} 1 \\ \lambda \\ 1 + \lambda \end{pmatrix} \quad \vec{PJ} = \begin{pmatrix} 1 - 1 \\ \lambda - 2 \\ 1 + \lambda - 0 \end{pmatrix} = \begin{pmatrix} 0 \\ \lambda - 2 \\ 1 + \lambda \end{pmatrix}$$

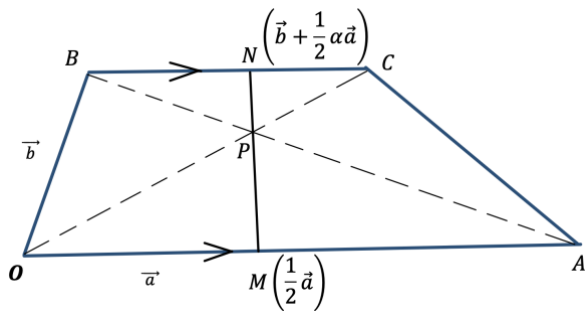
$$\begin{pmatrix} 0 \\ \lambda - 2 \\ 1 + \lambda \end{pmatrix} \cdot \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} = 0. \text{ (perpendicular)}$$

$$\lambda = \frac{1}{2}$$

$$\therefore \vec{PJ} = \begin{pmatrix} 0 \\ -3/2 \\ 3/2 \end{pmatrix}$$

$$|\vec{PJ}| = \sqrt{\left(\frac{3}{2}\right)^2 + \left(\frac{3}{2}\right)^2} = \frac{3}{\sqrt{2}}$$

d)
(i)



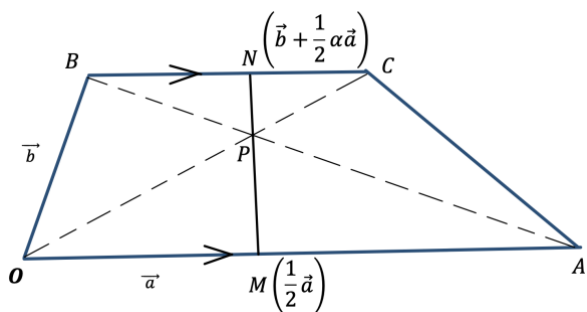
Since, $BC \parallel OA$, $\vec{BC} = \alpha \vec{OA} = \alpha \vec{a}$ for some constant α .

Equation of OC is

$$\vec{r}_1 = t (b + \alpha \vec{a})$$

1 mark: Explains that
 $\vec{BC} = \alpha \vec{OA}$ and
writes the vector
representations for OC

(ii)



Similarly, equation of AB is

$$\vec{r}_2 = \vec{a} + \lambda (\vec{b} - \vec{a})$$

Let P be the point of intersection of OC and AB .

Then, at point P ,

$$t(b + \alpha \vec{a}) = \vec{a} + \lambda (\vec{b} - \vec{a}) \text{ for some value of } t \text{ and } \lambda.$$

$\Rightarrow$

$$(t\alpha - 1 + \lambda)\vec{a} = (\lambda - t)\vec{b}$$

Since, $\vec{a}$ and $\vec{b}$ are non-parallel vectors, we must have

$$t\alpha - 1 + \lambda = 0 \text{ and } \lambda - t = 0$$

$$\lambda = t$$

$$\therefore t\alpha - 1 + t = 0$$

$$t = \frac{1}{\alpha + 1}$$

Hence, the position vector of P is:

$$\vec{r}_3 = \frac{1}{\alpha + 1} (\vec{b} + \alpha \vec{a}) \quad \text{1 mark}$$

Equation of MN :

$$r_4 = \frac{1}{2}\vec{a} + k \overrightarrow{MN}$$

$$= \frac{1}{2}\vec{a} + k (\vec{b} + \frac{1}{2}\alpha\vec{a} - \frac{1}{2}\vec{a})$$

1 mark: Equates the equations for OC and AB to help find B and gives

$$t = \frac{1}{\alpha + 1}$$

1 mark: compare coefficients and finds the position vector of P

1 mark: writes the vector equation of MN

1 mark: Proves that P lies on MN .

$$= \frac{1}{2} \vec{a} + k \left(\vec{b} + \frac{1}{2}(\alpha - 1)\vec{a} \right) \text{ 1 mark}$$

Note: Comparing r_3 and r_4 , we note that the coefficient $\vec{b}$ is

$$\frac{1}{\alpha+1}$$

Hence, k must be $\frac{1}{\alpha+1}$

$$\begin{aligned} r_4 &= \frac{1}{2} \vec{a} + \frac{1}{\alpha+1} \left(\vec{b} + \frac{1}{2}(\alpha - 1)\vec{a} \right) \\ &= \frac{1}{\alpha+1} \vec{b} + \frac{\alpha-1}{2(\alpha+1)} \vec{a} + \frac{1}{2} \vec{a} \end{aligned}$$

$$= \frac{1}{\alpha+1} \vec{b} + \frac{1}{2(\alpha+1)} (\alpha - 1 + \alpha + 1) \vec{a}$$

$$= \frac{1}{\alpha+1} (\vec{b} + \alpha \vec{a}) \text{ which is the position vector of P.}$$

That is P lies on MN **1 Mark**

Question 16

a)	Suppose c is a real number such that $r_3x^3 + r_2x^2 + r_1x + r_0 = 0$ where, r_0, r_1, r_2 and r_3 are rational numbers. By the definition of rational, $r_0 = \frac{a_0}{b_0}, r_1 = \frac{a_1}{b_1}, r_2 = \frac{a_2}{b_2} \text{ and } r_3 = \frac{a_3}{b_3} \text{ for some integers}$ 1 mark a_0, a_1, a_2 and $a_3 \in \mathcal{N}$ and non-zero integers b_0, b_1, b_2 and $b_3 \in \mathbb{Z}^+$. 1 mark By substitution, $r_3c^3 + r_2c^2 + r_1c + r_0 = \frac{a_3}{b_3}c^3 + \frac{a_2}{b_2}c^2 + \frac{a_1}{b_1}c + \frac{a_0}{b_0}$ $= \frac{a_3b_2b_1b_0}{b_3b_2b_1b_0}c^3 + \frac{a_2b_3b_1b_0}{b_3b_2b_1b_0}c^2 + \frac{a_1b_3b_2b_0}{b_3b_2b_1b_0}c + \frac{a_0b_3b_2b_1}{b_3b_2b_1b_0} = 0$ $b_3b_2b_1b_0 \neq 0$, by definition. 1 mark Multiplying by $b_3b_2b_1b_0$ gives $a_3b_2b_1b_0 \cdot c^3 + a_2b_3b_1b_0 \cdot c^2 + a_1b_3b_2b_0 \cdot c + a_0b_3b_2b_1 = 0$ Let $n_3 = a_3b_2b_1b_0, n_2 = a_2b_3b_1b_0, n_1 = a_1b_3b_2b_0$ and $n_0 = a_0b_3b_2b_1$ where n_3, n_2, n_1 and n_0 are all integers, being product of integers. Thus, c satisfies a polynomial with integer coefficients.	1 mark: expresses r_0, r_1, r_2 and r_3 as rational numbers 1 mark: clearly explains the domain of numerator and denominator of the rational coefficients. 2 marks: correct proof • Clear reasoning • Clearly demonstrates how the integer coefficients are derived 1 mark: attempts to prove that the coefficients are integers, however, gives clear reasoning for each assertion.	
(b)	$P(x(\theta)) = h + x(\theta)\tan\theta - \frac{g(x(\theta))^2}{2v^2}\sec^2\theta$		

	We are looking for the maximum value of $x(\theta)$, hence $x'(\theta) = 0$ $P'(x(\theta)) \cdot x'(\theta) =$ $x(\theta) \sec^2 \theta + x'(\theta) \tan \theta$ $-\frac{g}{2v^2} [(x(\theta))^2 \times 2 \sec \theta \cdot \sec \theta \tan \theta + 2x(\theta)x'(\theta) \sec^2 \theta]$ Put $x'(\theta) = 0$ $0 = x(\theta) \sec^2 \theta - \frac{g}{2v^2} (x(\theta))^2 \sec^2 \theta \tan \theta$ $\theta \neq \frac{\pi}{2}, x(\theta) \neq 0$ Hence, $x(\theta) \sec^2 \theta = \frac{g}{v^2} (x(\theta))^2 \sec^2 \theta \tan \theta$ $\frac{g}{2v^2} x(\theta) \tan \theta = 1$ $x(\theta) = \frac{v^2}{g} \cot \theta$ The given function is a concave down quadratic in $x(\theta)$. Hence, $x(\theta)$ will be maximum, when $\theta = \theta_m$ $x_m = \frac{v^2}{g} \cot \theta_m$	1 mark: differentiates the compound function 1 mark: sets $x'(\theta) = 0$ and solves for $x(\theta)$ to prove the result	
c(i)	$x = v \cos \theta t$ $\therefore t = \frac{x}{v \cos \theta}$ $y = h + v \sin \theta t - \frac{1}{2} g t^2$ $y = h + v \sin \theta \cdot \frac{x}{v \cos \theta} - \frac{1}{2} g \left(\frac{x}{v \cos \theta} \right)^2$	1 mark: eliminates the parameter θ from the two simultaneous equations of motion	

	$\therefore y = h + x \tan \theta - \frac{gx^2}{2v^2} \sec^2 \theta$		
c(ii)	An enveloping parabola can be got by maximizing the height of the projectile for a given horizontal distance x, which will give us the path that encloses all possible paths. $y = h + x \tan \theta - \frac{gx^2}{2v^2} \sec^2 \theta$ Let $u = \tan \theta$ $y = h + ux - \frac{gx^2}{2v^2} (1 + u^2)$ $\frac{dy}{du} = x - \frac{gx^2}{2v^2} \times 2u$ $\frac{dy}{du} = 0$ $x \neq 0$ $x = \frac{gx^2}{2v^2} \times 2u \Rightarrow u = \frac{v^2}{gx} \quad \mathbf{1 \text{ mark}}$ Hence, the envelope is $y_{\max} = h + \left(\frac{v^2}{gx}\right)x - \frac{gx^2}{2v^2} \left(1 + \left(\frac{v^2}{gx}\right)^2\right)$ $= h + \frac{v^2}{g} - \frac{gx^2}{2v^2} - \frac{gx^2}{2v^2} \times \frac{v^4}{g^2 x^2}$ $= h + \frac{v^2}{g} - \frac{gx^2}{2v^2} - \frac{gx^2}{2v^2} \times \frac{v^4}{g^2 x^2}$ Hence, the equation of the envelope is $\phi(x) = h + \frac{v^2}{2g} - \frac{gx^2}{2v^2}$	1 mark: Differentiates $y = h + ux - \frac{gx^2}{2v^2} (1 + u^2)$ and finds $\tan \theta$ that maximises y. 1 mark: substitutes u and gets the equation of the envelope.	

c(iii)	At the point $x = c$, $P(x)$ and $\phi(x)$ intersect the plane $y = mx$. $h + \frac{v^2}{2g} - \frac{gc^2}{2v^2} = mc$ $\frac{gc^2}{2v^2} + mc - \left(\frac{2gh + v^2}{2g}\right) = 0$ $c^2 + \frac{2v^2}{g} mc - \frac{2v^2}{g} \left(\frac{2gh + v^2}{2g}\right) = 0$ $c^2 + \frac{2mv^2}{g} c - \frac{v^2}{g^2} (2gh + v^2) = 0 \quad \mathbf{1 \text{ mark}}$ Hence, the value of c is $c = \frac{-\frac{2mv^2}{g} \pm \sqrt{\frac{4m^2v^4}{g^2} + \frac{4v^2}{g^2} (2gh + v^2)}}{2}$ $c = -\frac{mv^2}{g} \pm \frac{v^2}{g} \sqrt{m^2 + \frac{2gh + v^2}{v^2}}$ $c = -\frac{mv^2}{g} \pm \frac{v^2}{g} \sqrt{m^2 + \frac{2gh}{v^2} + 1} \quad \mathbf{1 \text{ mark}}$	2 marks: Equates the two functions and forms the simultaneous equation 1 mark: correctly solves to give the value of c, the x-coordinate of the point of impact.	
c(iv)	At the point $x = c$, $P(x)$ and $\phi(x)$ intersect the plane $y = mx$. That is $c = x_m$.	3 marks: refers to the solution from (b), sets $c = x_m$ and uses the result in (ii) to prove the result, with reasoning	

From (b), the optimal launching angle θ_m that gives the maximum range x_m is:

$$x_m = \frac{v^2}{g} \cot \theta_m \quad \mathbf{1 \text{ mark}}$$

Hence, from (ii),

$$c = x_m$$

$$-\frac{mv^2}{g} \pm \frac{v^2}{g} \sqrt{m^2 + \frac{2gh}{v^2} + 1} = \frac{v^2}{g} \cot \theta_m$$

$$\cot \theta_m = \frac{g}{v^2} \left(-\frac{mv^2}{g} \pm \frac{v^2}{g} \sqrt{m^2 + \frac{2gh}{v^2} + 1} \right)$$

$$\cot \theta_m = -m \pm \sqrt{m^2 + \frac{2gh}{v^2} + 1}$$

Optimal angle of projection θ_m

$$\theta_m = \cot^{-1} \left(-m \pm \sqrt{m^2 + \frac{2gh}{v^2} + 1} \right)$$

However, $0 < \theta < \frac{\pi}{2}$

Hence, $\cot \theta_m > 0$. and reject

2 marks: refers to the solution from (c), sets $c = x_m$ and uses the result in (ii) to prove the result

1 mark: refers to the result in (b) and identifies

$c = x_m$ and attempts to solve

	$\cot \theta_m = -m - \sqrt{m^2 + \frac{2gh}{v^2} + 1} < 0$		
	Hence, $\cot \theta_m = -m + \sqrt{m^2 + \frac{2gh}{v^2} + 1}$		